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What is Driving Reserve Accumulation? A Dynamic Panel Data Approach

*Diego Bastourre, Jorge Carrera, and Javier Ibarlucia**

Abstract

Foreign reserve accumulation is a widespread phenomenon of recent years, particularly among emerging economies. Using a panel of 136 countries for the period 1973–2003, we demonstrate the need of using both a dynamic specification of the reserve demand equation and the system GMM estimator. These improvements change the results found by previous studies. Openness, regional imitation, persistence, an inverted U-shaped relationship between reserves and income level, and financial deregulation are the factors driving reserve hoardings. In contrast, reserves yield, and both trade and financial volatility are not statistically significant. Surprisingly, we find that countries with flexible exchange rate regimes have higher ratios of reserves to GDP. This result is robust to alternative exchange rate regime classifications.

1. Introduction

The strong increase in international reserves is one of the most surprising and controversial recent issues in the field of international macroeconomics. The interest in the causes and consequences of this phenomenon are obvious once the data are observed. In fact, in the period 1995–2005 foreign reserves increased by 180%, reaching a world record of US\$4.3 trillion (see Figure 1).¹

Accumulation is widespread across countries even when the world stock of reserves is strongly influenced by some well-known players such as China, Japan, India, or Russia. However, the pace of hoarding is neither homogeneous across countries nor over time, as can be seen in Table 1. It shows the annual growth rates of foreign reserves in the post-Bretton Woods (BW) era splitting a sample of 139 countries in two periods (1973–90 and 1990–2004), and grouping by three criteria (OECD vs emerging markets, income level, and geographic region). Emerging economies of South and East Asia are the current leaders in the reserve accumulation race. In contrast, Europe and the US have been falling behind.

As surprising as these growth rates is the observed heterogeneity in reserve levels. As stylized evidence of this fact, the evolution of the ratio reserves to GDP by region during 1973–2003 is summarized in Table 2. The general picture that emerges from the data has raised new questions to be answered by the international macroeconomics literature. Why do so many countries accumulate international reserves? Is there a

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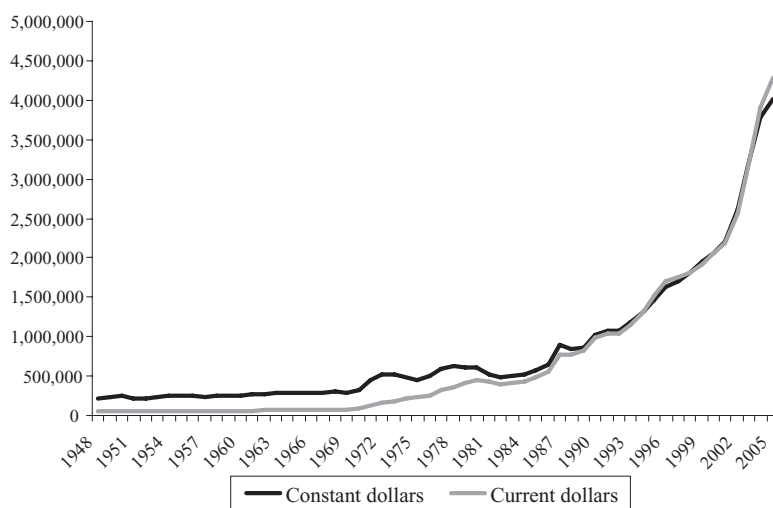


Figure 1. *Total International Reserves, 1948–2005 (millions of dollars)*

Table 1. *International Reserves Annual Growth Rates, 1973–2004 (per cent)*

Country group	Period	
	1973–90	1990–2004
<i>Markets</i>		
OECD	3.09	5.27
Emerging	4.20	17.36
Others	3.94	9.84
<i>Income level</i>		
Low	−0.30	21.57
Lower Middle	3.22	18.14
Upper Middle	4.07	10.27
High Non-OECD	6.76	9.66
High OECD	3.19	6.10
<i>Regions</i>		
East Asia & Pacific	6.15	16.77
South Asia	−2.52	29.74
Europe and Central Asia	2.80	0.28
Latin America & Caribbean	1.20	10.71
North America	3.69	0.69
Middle East & North Africa	1.89	11.51
Sub-Saharan Africa	1.63	10.86

unique motivation for stockpiling reserves? What are their roles in an era of capital liberalization and exchange rate flexibility? Are the theoretical models and the empirical estimations so far adequate to explain the central banks' behavior?

In this challenging research agenda, this paper deals with the issue of reserve determinants using an econometric approach. We employ large and time-extended panel

Table 2. Reserves to GDP Ratio by Regions, 1973–2003 (per cent)

Country group	Period	
	1973–90	1990–2003
<i>Regions</i>		
East Asia & Pacific	3.65	8.89
South Asia	2.81	6.22
Europe and Central Asia	5.79	5.01
Latin America & Caribbean	5.09	7.80
North America	1.08	1.16
Middle East & North Africa	15.72	13.73
Sub-Saharan Africa	4.48	7.50

data to test the economic factors explaining the observed heterogeneity in reserve levels.

Our main contributions are two. The first one is testing the importance of variables which are supported on theoretical grounds but have not been considered so far: an inverted U-shaped income-level relationship, *de facto* exchange rate regimes, and herd (or imitation) behavior. These new determinants are added to the standard regression framework stressed by theoretical and empirical models. As a result, a more completed and balanced picture of the reasons to accumulate reserves in the current global context is obtained.

The second contribution is dynamically specifying the reserve demand equation. Although this is a natural way of modeling a stock demand, this fact has been neglected in recent empirical literature on this subject. This point seems somewhat technical at first sight, but is also significant because it changes the economic interpretation of the regression results. This, in turn, could mislead economic interpretations and policy recommendations. We also show there are costs of omitting reserves inertia, in terms of bias and consistency of the estimators. Nevertheless, incorporating dynamics carry some technical difficulties.

Traditional dynamic pooled ordinary least squares (OLS hereafter) and fixed effects estimators suffer from bias, so an alternative methodology is required. Several options are considered in this paper, but we conclude that the system GMM estimator is the most suitable one since it simultaneously solves the problems posed by the inclusion of inertia, the strong persistency of the reserve demand, and the potentially endogenous covariates.

The rest of the paper is organized as follows. Section 2 presents some theoretical considerations distinguishing the different branches in the reserve demand literature. Section 3 describes both the main drawbacks of previous empirical research on this field and the econometric methodology that solves these problems. The variables selected, their sources, and construction are explained in the fourth section; the significance of including new determinants not explored so far is also discussed. In the fifth part of the paper the empirical results are presented and discussed. Section 6 comprises the conclusion and policy implications suggested by the evidence.

2. Theoretical Considerations

The ideas regarding international reserves have changed over time. Traditional views emphasized the main role of three variables. The first one is the benefit of building up

reserve stocks that could avoid national income reduction to shrink imports. This is calculated by the reciprocal of the marginal propensity to import. The second factor is the opportunity cost of holding reserves, which is the spread between the interest rate earned by reserves and the alternative social use of these resources. The third traditional variable is the volatility of balance of payments, to take into consideration the degree of synchrony between external flows.

Pioneering work by Heller (1966) obtained an explicit formula for optimal reserves as a function of those factors, and assumed a probabilistic distribution of external deficits and surpluses in a cost–benefit framework. Subsequent papers by Clark (1970), Kelly (1970), Hamada and Ueda (1977), and Frenkel and Jovanovic (1981) modified some algebraic relationships and changed a few assumptions, but did not alter the fundamental conclusions: optimal reserves increase with volatility and decrease with propensity to import and the opportunity cost. Since it is connected to the methodological discussion in the next section it is important to note that this literature focuses on the stochastic time-series properties of reserves, highlighting the dynamic nature of reserve stock adjustments.

The Bretton Woods crisis was a major milestone in the international finance architecture that also challenged the established view regarding reserves. Previous to the BW breakdown, there was an important debate concerning liquidity creation that eventually gave rise to special drawing rights. The idea was to separate liquidity dependence from dollar injection originating in the United States' balance of payments deficits in order to avoid the Triffin paradox. Once the Smithsonian Agreement of 1972 finally crumpled and the main international currencies started to float in 1973, it was believed that international reserves were set to diminish. But contrary to this, accumulation began to grow fast under exchange rate flexibility. This puzzling outcome was first pointed out by Frenkel (1983). In spite of some explanations offered by Frenkel or Grimes (1993), it continues to be surprising to observe today this empirical regularity of increasing reserve stocks in a more flexible exchange rate regime environment.

However, to put this result in place it must be taken into consideration that the passage to flexible regimes was done in a period of capital account liberalization. Intuitively, this could offer a rationale for a shift in the demand of reserves. Precisely, the core of the most recent literature on foreign reserves views these assets as a precautionary cushion against sudden stops and financial volatility.

The precautionary approach has covered various related fields. In the first place, it is the issue of reserves as an insurance against sudden stops and capital reversals. Aizenman and Marion (2004) examined a two-period economy that experiences productivity shocks and can borrow abroad up to a limit due to the existence of some probability of default in the second period. Reserves provide insurance as long as there is a difference between first-period and second-period marginal utility, when default occurs, and there is no reserve cushion. International liquidity could help the economy to smooth consumption, even in the default scenario. Jeanne and Rancière (2006) have also worked in this field calibrating a model for optimal reserves in emerging countries. An important feature of this model is the different maturity of private (short) and public (long) finance. In the margin, reserves have to be financed with long-term debt since short positions are very difficult to roll over during a crisis.

In second place is the relationship between international reserves and currency crisis. In a second-generation crisis model, international liquidity could play a determinant role, capable of triggering a currency crisis when fundamentals are in an intermediate zone. In this line of work, Li and Rajan (2005) have formalized the idea that high reserves can help offset moderately weak fundamentals.

The precautionary approach also highlights a two-way relationship between reserves and financial risk. On the one hand, credit risk agencies monitor reserves, among other variables, in order to assign ratings to sovereign debt. Since such ratings influence investors' beliefs, there exists an important signaling effect of reserves. Because of the impossibility of establishing a unique adequacy criterion, the battle for reserves view (a semblance of the Keynes beauty contest) highlights that reserves should be similar among countries of similar characteristics to avoid a negative judgment of financial markets.

On the other hand, it is possible to argue that in a riskier environment, more reserves will be necessary. For instance, Ben-Bassat and Gottlieb (1992) developed a model to stress the cost of past default on the demand for international reserves. A default record would entail proportionally more reserves.

Besides precautionary motives, there is a branch in current international reserves literature that considers accumulation as a byproduct of real exchange rate targeting. This vision is mainly supported by Michael Dooley and Peter Garber (for a survey, see Dooley et al., 2005).

As we have seen from the figures in the introduction, the strong growth in international reserves and the heterogeneity in the reserves levels appear as uncontroversial facts. Conversely, the theoretical reasons explain that they are still a matter of debate (Aizenman and Lee, 2005, 2006). Therefore, a careful econometric analysis could be a suitable strategy to shed some light on the weights that correspond to each economic determinant.

3. Econometric Methodology

The use of a static reserve demand equation estimated by OLS (or generalized least squares, GLS) fixed effects has been common practice among empirical papers on this subject. This is the case, for instance, in the works of García (1999), Flood and Marion (2002), Aizenman and Marion (2003, 2004), Soto et al. (2004), or Aizenman and Lee (2005).

In using a static formulation these papers are implicitly assuming instantaneous adjustment to the equilibrium level of reserves when there is a shock in reserve determinants. However, this is not a realistic assumption.

Let us consider as an example an economy that is closed to international financial flows. At a given point in time this country has international reserves in accordance with its steady-state equilibrium represented by any standard formula (for instance, the formula in the model of Frenkel and Jovanovic, 1981). Suppose that at this point there is a sudden change in the institutional framework that aggressively opens the capital account. The central bank will need to equate its reserve level with the new long-run equilibrium. In a static formulation this is done entirely in a period of length equal to the frequency of the data (typically quarters or years). In the actual world, it could take several years to reach the new stock equilibrium considering the stochastic nature of the flows in the balance of payments and the side effects associated with the changes in the reserve stock.

If it is the case that reserve stocks change slowly from one year to another, then partial adjustments from current to desired reserve levels are the rule rather than the exception. The proper way to estimate this type of relationship should include the dynamic component in the international reserves demand (first lag of the dependent variable) to avoid misspecifications. This setting will allow us to make a distinction between short- and long-run elasticities.

Moreover, as noted by Bond (2002), even when the coefficient of lagged dependent variable could not be the focus of interest, allowing for dynamics in the underlying process may be crucial for recovering consistent estimates of other parameters, i.e. reserves demand determinants. But apart from this, existing empirical works of international reserve determinants have two additional drawbacks. On the one hand, they usually do not control for endogenous regressors in the right side of the demand equation. On the other hand, data coverage is limited.

The strategy of this paper is to employ the system GMM methodology (Arellano and Bover, 1995; Blundell and Bond, 1998) to deal with the issues of dynamic specification and endogeneity, and panel data containing 136 countries and 27 years to address the last point.

The inclusion of the lagged dependent variable in the demand equation causes direct endogeneity problems that need some treatment. Consider the following model:

$$y_{i,t} = \gamma y_{i,t-1} + x'_{i,t} \beta + \eta_i + \varepsilon_{i,t}, \quad (1)$$

where $x_{i,t}$ is a set of exogenous variables,² η_i is an individual effect that captures the unobserved heterogeneity among countries, and $\varepsilon_{i,t}$ is an unobserved homoskedastic disturbance term.³

There are several techniques to estimate a model like (1). One possibility is to apply OLS or GLS to a fixed effects model. However, the inclusion of $y_{i,t-1}$ downward biases the autoregressive coefficient since $y_{i,t-1} - \bar{y}_i$ is correlated with $\varepsilon_{it} - \bar{\varepsilon}_i$, as is well-documented by Nickell (1981) and Kiviet (1995). The alternative of a simple OLS estimation (ignoring individual heterogeneity) would produce an upward-biased autoregressive coefficient (Hsiao, 1986). The fact that the previous estimators are biased in opposite directions gives us a useful hint about the range in which the consistent estimation is expected to be.

An alternative procedure suggested by Anderson and Hsiao (1981) consists in differentiating equation (1), removing the η_i component to obtain:

$$(y_{i,t} - y_{i,t-1}) = \gamma(y_{i,t-1} - y_{i,t-2}) + (x_{i,t} - x_{i,t-1})' \beta + (\varepsilon_{i,t} - \varepsilon_{i,t-1}). \quad (2)$$

This transformation takes advantage of the absence of correlation between $(\varepsilon_{i,t} - \varepsilon_{i,t-1})$ and $y_{i,t-2}$ or $(y_{i,t-2} - y_{i,t-3})$, and hence the authors recommend using them as internal instruments for equation (2).

However, Anderson–Hsiao (1981) is a special case within the generalized method of moments (GMM) estimators (Hansen, 1982). More general GMM estimators, such as those proposed by Arellano and Bond (1991), are superior for at least two reasons. On the one hand, they gain efficiency by using as instruments all the available lags of the right-hand-side variables. On the other hand, they allow controlling for potential endogeneity of other variables different from $y_{i,t-1}$.⁴

Nevertheless, this methodology could suffer from small-sample bias due to the presence of weak instruments. As emphasized by Blundell and Bond (1998), Blundell et al. (2000), or Bond et al. (2001), lagged levels are weak instruments if the dependent variable is highly persistent, i.e. $\alpha \rightarrow 1$. To mitigate this problem it has been proposed to add an extra assumption, $E(\eta_i \Delta y_{i-2}) = 0$, which implies that there is no correlation between the error term $\varepsilon_{i,t}$ and $\Delta y_{i,t-1}$ for $t = 4, 5, \dots, T$.⁵ In this setting it is possible to estimate a level equation which is instrumented by properly lagged differences. The so-called system GMM estimator is a linear combination of the differences and levels estimators in which the weight of the latter increases as long as $\alpha \rightarrow 1$.

Summing up, the proper study of foreign reserves demand requires the simultaneous solution of the problems posed by adding dynamics, the strong persistency that potentially characterizes reserves and the endogenous $x_{i,t}$ regressors. From the previous discussion, the most suitable instrument to deal with these problems is the system GMM estimator; it will therefore be the preferred empirical specification throughout the paper.

4. Standard and New Determinants of Reserve Holdings

In this section we briefly describe the variables employed in our empirical model. Additional details concerning data sources and the construction of the variables are provided in the Appendix. The dependent variable is the log of the ratio of international reserves to GDP, as is common practice among empirical papers.

From the traditional literature we select three variables. First, the opportunity cost of holding international reserves is considered. Since it is very difficult to find a reliable rate of return for the alternative use of reserves for each country due to data imperfections, we only measure the proceeds earned by reserves with the yield of the 10-year US Treasury bond. Secondly, exports plus imports to GDP is included as a proxy of trade openness.⁶ Finally, we incorporate the volatility of commercial transactions calculated by the coefficient of variation of export levels using a five-year rolling window. We expect a positive sign for the coefficients of the reserves yield and trade openness variables, and a negative sign in the case of trade volatility.

Considering the insights of the general discussion of recent models on international reserves, we test three important factors: (i) the degree of openness of the capital account measured by volume of capital inflows relative to GDP; (ii) a volatility measure of these capitals inflows—constructed in the same way as trade volatility; and (iii) a dummy shift variable from 1990 to 2003 to capture the structural break regarding financial deregulation in a large number of developed and emerging countries. In all cases, we would expect to find positive coefficients in the regression analysis.

Up to this point we are describing variables that are standard in empirical research on international reserves. However, we think it is important to expand the control set to tackle three main issues: exchange rate regimes, development level, and regional imitation.

The exchange rate regime in place could influence the demand for international reserves. The standard belief is that fixed regimes need *ceteris paribus* more reserves. But as we mentioned in the second section, this was not verified in the data. Frenkel's (1983) explanation for this paradox was that central banks' behavior did not substantially change after BW breakdown. Many of the exchange rate arrangements during BW were not exchange rate pegs but adjustable parities. Conversely, after the 1972–73 crisis, central banks carried out managed floating rather than pure floating. To test the effect of the exchange rate regime on the reserve holdings avoiding the Frenkel critique we use a *de facto* exchange rate regime classification. Specifically, we have extended the classification developed by Coudert and Dubert (2005) to cover 136 countries and 31 years of our dataset.

The classification is based on three steps. The first one consists of the estimation of a trend in the (monthly) nominal exchange rate in order to distinguish crawling peg from peg regimes. The second stage makes it possible to separate fixed regimes (pegs and crawling peg) from flexible ones (managed and pure floating). This is done by constructing a comparison test of annual nominal exchange rate variance between each

country and a benchmark group of floating currencies (pound sterling, Deutschmark, and yen) against the dollar. In the third step, a comparison test is performed on the variance of monthly changes of international reserves against the benchmark group in order to differentiate managed from pure floating.

Finally, we have constructed three categories: fixed regimes (peg and crawling peg), flexible regimes (floating and managed floating) and inconclusive (due to missing data). Another contribution is the potential inverted U-shaped relationship between reserves holdings and economic development.

Since middle-income countries are more prone to financial crisis than rich or poor countries, they are more interested in using reserves as a precautionary buffer against sudden stops and financial volatility. These countries do not have a fully developed institutional framework to deal with external shocks such as developed countries have. They usually experience fear of floating and cannot rest only on exchange rate flexibility as a shock absorber. Additionally, they do not have full access to international credit markets when a shock hits the economy and cannot count on a reliable international lender of last resort. Besides, poor countries are not financially integrated and have far greater priorities than stockpiling reserves, even if we assume they have the ability to generate regular balance of payments surpluses. That is, the social return of an alternative use of international reserves is much higher than in middle-income countries. For these reasons we have incorporated into our regression analysis the level of the GDP PPP per capita in constant dollars and its quadratic term.

The last new variable that we incorporate in the regression analysis is a measure of the degree of regional imitation. There are at least two reasons to mimic the actions of neighbors of similar characteristics based on strategic behavior. On the one hand, it can be asserted that there exists a relationship between reserves and risk. A country with more reserves is perceived to be less risky than a neighbor by credit-rating agencies (and so by the international financial community in general) all things being equal. It is possible to imagine a Nash equilibrium characterized by high reserve stocks, in which each country tries to increase its reserves relative to similar countries to reduce perceived risk, but finally none succeeds in effectively reducing it. This is the idea of the battle of reserves examined in the theoretical section.

On the other hand, it is possible to find a Nash-equilibrium scenario with high levels of reserves in those regions pursuing export-led growth policies. Some countries might be reluctant to lose competitiveness, and thus they try to avoid real appreciation *vis-à-vis* their neighbors. To this end, central banks of those countries probably have to stockpile reserves and maintain some type of capital controls.⁷ The variable included to capture the degree of regional imitation is the percentage of countries of each geographical region that increased their reserves in the previous year.

5. Empirical Results

As we have previously discussed, the proper identification of reserve drivers requires a dynamic setting. To this end, the methodological options are presented in the first part of this section. After corroborating the empirical relevance of inertia and its consequences on the estimation technique, we move on to the analysis of the results of our preferred system GMM model. In the last part of this section we present some robustness checks considering alternative exchange rate classifications, as well as *de jure* measures of capital account openness.

Static vs Dynamic Reserve Demand Specification

We start estimating a static fixed effects model with the objective of establishing a benchmark case with similar characteristics to those found in several papers on this subject. This is done in the first column of Table 3. In the remaining columns of Table 3 we estimate various dynamic models. Columns (2) and (3) are the fixed effects and pooled OLS dynamic models (D-LSDV and pooled DOLS, respectively). These two methodologies produce biased (in opposite directions) autoregressive coefficients. For this reason they are useful as a guide providing a range for a consistent estimation of the autoregressive parameter. We observe in Table 3 that the lower and upper limits of this interval are 0.7325 (D-LSDV) and 0.8923 (pooled DOLS), respectively. It is not

Table 3. Econometric Estimations

	<i>LSDV</i>	<i>Pooled DOLS</i>	<i>D-LSDV</i>	<i>Diff. GMM</i>	<i>System GMM</i>
<i>Variables</i>	(1)	(2)	(3)	(4)	(5)
Inertia		0.8923*** [0.000]	0.7325*** [0.000]	0.6181*** [0.000]	0.7602*** [0.000]
GDP PPP PC	0.00004249** [0.045]	0.000007857** [0.022]	-0.00002540** [0.032]	0.00005393 [0.281]	0.00003506* [0.084]
GDP PPP PC sq.	-2.462e-09*** [0.000]	-4.384e-10*** [0.000]	-1.916e-10 [0.424]	-2.169e-09* [0.054]	-1.911e-09*** [0.009]
Trade Openness	0.6515*** [0.000]	0.07630*** [0.002]	0.2614*** [0.000]	0.4078** [0.016]	0.4963*** [0.000]
Financial Openness	0.02934 [0.200]	0.01314 [0.361]	0.03126* [0.056]	0.02629 [0.354]	0.05442** [0.016]
Trade Volatility	0.07543*** [0.003]	-0.01180 [0.525]	-0.009612 [0.553]	-0.04367** [0.028]	-0.02230 [0.336]
Financial Volatility	-0.08744** [0.011]	0.03078 [0.112]	0.02573 [0.249]	0.03824 [0.362]	0.002159 [0.945]
Flexible ERR	-0.07041** [0.027]	0.02338 [0.169]	0.04138* [0.057]	0.04791** [0.036]	0.06441** [0.031]
Inconclusive ERR	-0.4203*** [0.005]	0.003570 [0.974]	-0.04481 [0.675]	0.03810 [0.741]	0.0009314 [0.991]
Reserves Yield	-0.6011*** [0.000]	0.01139 [0.815]	-0.1006* [0.061]	-0.1202* [0.075]	-0.03591 [0.570]
Dummy Shift 1990	0.1813*** [0.000]	0.1099*** [0.001]	0.1301*** [0.000]	0.07691* [0.074]	0.08703** [0.011]
Regional Imitation	0.001505* [0.071]	0.003956*** [0.000]	0.003525*** [0.000]	0.003160*** [0.000]	0.004001*** [0.000]
Constant	-1.2108*** [0.000]	-0.5477*** [0.000]	-0.3669* [0.053]		-0.5819*** [0.004]
Countries	136	136	136	136	136
Observations	2327	2321	2321	2161	2321
F-test	0.000***	0.000***	0.000***	0.000***	0.000***
No. of instruments	—	—	—	129	129
Hansen test	—	—	—	0.484	0.450
Diff. Hansen test	—	—	—	—	0.258
AB test for AR(1)	—	—	—	0.000***	0.000***
AB test for AR(2)	—	—	—	0.434	0.342

Notes: *p*-Values are in square brackets. Significance levels: * 10%, ** 5%, *** 1%.

necessary to go through more sophisticated models to verify a high degree of persistence in the demand of international reserves.

Then, we turn to GMM family estimators that allow a consistent dynamic estimation in the presence of endogenous x_{it} variables. Column (4) of Table 3 is the traditional difference estimator of Arellano and Bond (1991).⁸ Regarding endogenous variables, we have instrumented the lagged dependent variable, the GDP PPP per capita level and its square, as well as both trade and financial openness variables. The results of both the Hansen test of overidentifying restrictions and the second-order residual autocorrelation test do not reject the validity of the instrumental set.

In the differences GMM model, the coefficient of lagged reserves to GDP ratio is lower than the value of the downward-biased D-LSDV estimate and the difference is quite significant (0.6181 vs 0.7325). As was mentioned in the methodological section this fact is a clear signal of biased coefficients caused by weak instruments.

System GMM estimates are in column (5). As can be seen, the autoregressive coefficient of the lagged dependent variable increases considerably (0.7602) and lies between the pooled OLS and D-LSDV models. As in the previous case, the Hansen and the second-order autocorrelation tests do not reject the validity of the instrumental set.⁹ Moreover, the difference Sargan test does not reject the null hypothesis of valid additional moment conditions imposed in the system estimator with respect to the difference estimator.

Hence, we have confirmed that the system GMM could deal with high persistency proper of reserves stock demand. The economic interpretation that we previously offered dates back to the pioneering first-generation models of reserves adequacy that stressed stochastic time-series properties of reserves adjustments.¹⁰ Since we have solved the technical issues posed by the dynamic demand reserve estimation, we will turn next to the analysis of the statistical and economic significance of each determinant.

Analysis of the Reserve Determinants

Concerning reserve determinants, our preferred system GMM estimates—column (5) in Table 3—suggest many motivating results.

Regarding traditional explanatory variables, only trade openness is statistically significant with the expected positive sign. Neither the reserves yield nor the trade volatility seems to be significant at conventional levels. In the former, the lack of significance could indicate that the available information does not allow us to construct a good proxy of the opportunity cost.¹¹ Besides, as has been argued by Rodrik (2006), the relevant opportunity cost is not as simple as an interest rate differential. Benefits could include both the reduction in the likelihood of financial crisis and the size of the adjustment when the crisis occurs. Reserves could also cause positive externalities for the private sector in the form of more and cheaper finance. Obviously, all these factors are not easy to quantify.

Interestingly, the financial opening variable is statistically significant, and the corresponding financial flows volatility is not. We suppose this important effect is partially captured by the dummy variable for the period 1990–2003, an era characterized by worldwide capital account deregulation (Chinn and Ito, 2007). In conjunction, the results of trade and financial openness, and trade and financial volatility, seem to indicate that, for reserve accumulation purposes, it is more important to cover the average shock exposure rather than its variance.

We find strong evidence that the new proposed determinants are relevant. An inverted U-shaped relationship is verified between reserves and the development level

measured by GDP PPP per capita. We have proposed some reasons for this singular functional form in the previous section. Using the coefficients of the linear and quadratic GDP PPP per capita terms from column (5), the expression for this nonlinear relationship is:¹²

$$\log\left(\frac{\text{Reserves}}{\text{GDP}}\right) = (-1.91 * 10^{-9}) * \text{GDP}_{pc}^2 + (3.506 * 10^{-5}) * \text{GDP}_{pc} + c, \quad (3)$$

where c is a constant that includes the effects of the other explanatory variables at a given point. Hence, *ceteris paribus* the ratio of reserves to GDP has a maximum when GDP per capita is:

$$\text{GDP}_{pc}^{MAX} = \frac{(3.506 * 10^{-5})}{2 * (1.91 * 10^{-9})} = 9172. \quad (4)$$

To put this value in context, consider that the mean income in the sample is 8445 PPP dollars at 2000 constant prices. Additionally, 66.5% (33.5%) of the included countries shows an income that is below (above) this figure. Among several representative middle-income countries that exhibited a similar income level we find, for instance, the cases of Hong Kong in 1976, Portugal (1977), Singapore (1979), Argentina (1990), South Africa (1993), Malaysia (1997), Chile (1999), Mexico (2000), or Russia (2003).

The nonlinear relationship is depicted in Figure 2. To construct it we solve first the long-run equilibrium ($y_t = y_{t-1}$). Thus, long-run elasticities are obtained multiplying the impact coefficients of explanatory variables by $1/(1 - \alpha)$. After that, it is necessary to define a value for the constant c of equation (3). As an example we consider the case in which all explanatory variables are set in their sample mean.¹³ In a final step, we recover the ratio of reserves to GDP. All in all, this curve reinforces the idea that middle-income countries are those more willing to pursue aggressive accumulation strategies.

Regarding the regional imitation variable, both the sign and the statistical significance of the proxy suggest that reserve accumulation strategies could be considered a non-cooperative game. In this sense, it seems that each player of the reserves game is trying to not fall behind competitors in terms of international risk perception and/or real exchange rate competitiveness.

The last point focuses on the exchange rate regime variables, which leaves us with a more puzzling result. We find that, other things constant, flexible regimes have a ratio of reserves to GDP 6% higher than fixed exchange rate regimes (the latter is the

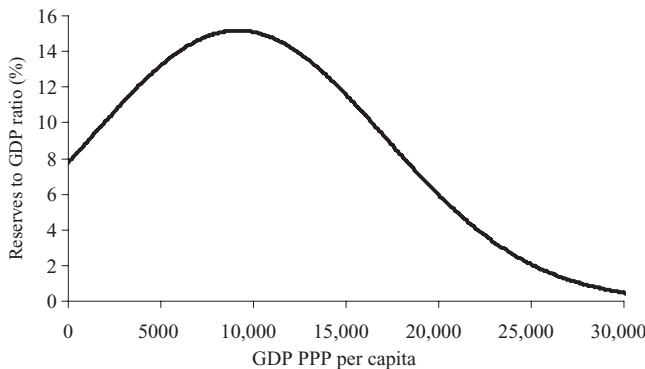


Figure 2. The Implicit Relationship between Reserves and Income Level

omitted category). As we have explained, this result is not subject to the Frenkel critique because we are employing a *de facto* classification.

We interpret this result as induced by the behavior of policymakers who are willing to float only when they can show that they have the ability to intervene and reduce nominal exchange rate volatility. In a kind of corollary of the Calvo and Reinhart (2002) hypothesis, we could argue that developing economies usually experience fear of floating, but when they float, it is because they are (or at least they think they are) able to fix (if they want). In some way, this means that policymakers of emerging economies are stockpiling reserves and allowing exchange rate fluctuations as a complementary strategy, in order to minimize volatility and reduce crisis probability.

Robustness Checks

To check the robustness of the results we perform additional estimations employing alternative proxies for some explanatory variables. For reasons already discussed, all the regressions in this part are performed with the system GMM methodology.

Certainly, the surprising finding about exchange rate arrangements deserves a more careful examination. A possible concern could be the classification of regimes per se, since this is an issue that is not completely settled in the literature. Hence, as a first robustness check we consider the *de facto* classification of Levy-Yeyati and Sturzenegger (2005), grouping it in fixed (omitted category), flexible, and inconclusive regimes.

Results are shown in the first column of Table 4. Once again, flexible arrangements entail more reserves since the respective dummy variable results are positive and significant at the 5% level. Notably, the remaining determinants are almost unchanged in terms of their statistical relevance and economic magnitude.

But some caveats may still persist. For instance, it could be argued that the results are driven by countries pursuing strategies connected to real exchange rate level and/or volatility. This line of reasoning would stress a difference between managed and pure floating. A way of controlling for this type of effect is employing all the categories of both the Coudert and Dubert (2005) and Levy-Yeyati and Sturzenegger (2005) classifications; splitting fixed regimes in pegs and crawling peg, and flexible regimes in floating and managed floating.

The results of the “fine” Coudert and Dubert classification and the “fine” Levy-Yeyati and Sturzenegger classification are reported in columns (2) and (3) of Table 4, respectively. In both cases the categories pure floating, managed floating, and crawling peg show positive coefficients which are statistically significant at the 5% level. This confirms that the result is not exclusively driven by countries that practice managed floating. However, there are some differences between specifications. For instance, the highest coefficients among these dummies are for the pure floating category in Coudert and Dubert and for the managed floating dummy if the methodology of Levy-Yeyati and Sturzenegger is employed. Further research on this topic could explore possible interactions of exchange rate arrangements with other characteristics of the macroeconomic regime, such as monetary, fiscal, commercial, and development policies.

Since current models and discussions about reserve accumulation have emphasized the role of capital account deregulation, it could be useful to incorporate an explicit role for capital account controls. In this sense, our financial openness variable proxies the *de facto* integration, but there are several cases in which the observed flows do not necessarily concur with actual restrictions on capital account movements.

Two additional regressions are carried out employing the Chinn and Ito (2007) *de jure* measure of capital controls (*KAOPEN*), which is particularly useful to analyze

Table 4. Robustness Checks of the Econometric Results

	<i>Collapsed LY-S classification</i>	<i>Fine C-D classification</i>	<i>Fine LY-S classification</i>	<i>KAOPEN measure</i>	<i>KAOPEN measure</i>
<i>Variables</i>	(1)	(2)	(3)	(4)	(5)
Inertia	0.7757*** [0.000]	0.7617*** [0.000]	0.7798*** [0.000]	0.7557*** [0.000]	0.7594*** [0.000]
GDP PPP PC	0.0000324* [0.054]	0.0000298 [0.124]	0.0000286* [0.095]	0.0000289 [0.175]	0.0000410* [0.065]
GDP PPP PC sq.	-1.65e-09*** [0.0070]	-1.70e-09** [0.014]	-1.47e-09** [0.017]	-1.60e-09** [0.037]	-2.08e-09*** [0.005]
Trade Openness	0.4856*** [0.000]	0.4824*** [0.001]	0.4471*** [0.000]	0.5846*** [0.000]	0.4642*** [0.002]
Financial Openness	0.0638** [0.026]	0.0561** [0.011]	0.0588* [0.031]		0.0562** [0.014]
<i>De Jure</i> Financial Openness				0.0186 [0.456]	0.0188 [0.484]
Trade Volatility	-0.0290 [0.306]	-0.0243 [0.286]	-0.0285 [0.288]	-0.0152 [0.522]	-0.0167 [0.482]
Financial Volatility	0.0230 [0.465]	0.0145 [0.644]	0.0248 [0.410]	0.0190 [0.572]	0.0145 [0.637]
Flexible ERR	0.1267** [0.012]			0.0644* [0.073]	0.0540* [0.086]
Pure Floating		0.1575*** [0.001]	0.1217** [0.041]		
Managed Floating		0.0920** [0.016]	0.2280** [0.001]		
Crawling Peg		0.0967** [0.024]	0.1272** [0.024]		
Inconclusive ERR	0.1545** [0.0034]	0.3935 [0.654]	0.1775** [0.034]	0.0088 [0.920]	0.0291 [0.742]
Reserves Yield	-0.0036 [0.960]	-0.0278 [0.663]	-0.0010 [0.989]	-0.0192 [0.785]	-0.0248 [0.683]
Dummy Shift 1990	0.0541 [0.146]	0.0837** [0.013]	0.0614 [0.112]	0.0789** [0.034]	0.0880** [0.019]
Regional Imitation	0.0040*** [0.000]	0.0041*** [0.000]	0.0038*** [0.000]	0.0040*** [0.000]	0.0040*** [0.000]
Constant	-0.5782*** [0.008]	-0.6294*** [0.002]	-0.6178*** [0.006]	-0.7190*** [0.000]	-0.6104*** [0.002]
Countries	132	136	132	136	136
Observations	2000	2321	2000	2265	2265
No. of instruments	129	131	131	99	130
Hansen test	0.440	0.504	0.435	0.342	0.385
AB test for AR(2)	0.226	0.343	0.215	0.332	0.402

Notes: *p*-Values are in square brackets. Significance levels: * 10%, ** 5%, *** 1%.

the extension of restrictions. In one of these regressions the financial openness variable is replaced by *KAOPEN* and, in the other, they are jointly considered.

In both specifications this *de jure* measure does not result from a statistical point of view, although its sign is positive as expected. However, as could be observed in column (5) of Table 4, not only the *de facto* variable is robust to the inclusion of *KAOPEN* but also the rest of the traditional and novel determinants studied in this paper.

6. Conclusions

Since the 1990s, the interest in international reserves has been growing rapidly, but the established view nowadays is rather different from the prevailing ideas of the 1960–80 period. The main reason to demand international liquidity is basically the same: precaution. However, what has probably been changing is the source of uncertainty that motivates precautionary behavior: from protection from current account disturbances, to capital account shocks hedging.

This paper recovers the spirit of traditional models which highlight the adjustments cost of changing reserve stocks from current to desired levels. This recognition naturally leads to a framework, characterized by persistence in time series. We have demonstrated that ignoring reserve demand dynamics is a potential flaw of recent empirical papers, which tends to be intensified in high persistency scenarios.

But adding dynamics is costly. A proper study of foreign reserves demand requires solving two problems simultaneously: the strong persistency that characterizes reserves, and the potentially endogenous $x_{i,t}$ regressors. According to our evaluation the most suitable instrument to address these problems is the system GMM estimator.

In order to specifically answer why countries accumulate reserves, we have proposed a series of traditional and new variables supported in classical and modern theoretical grounds. We confirm that the only traditional variable that remains significant is the trade openness, while neither trade and financial volatility nor opportunity cost is relevant. Other significant variables are financial openness, regional imitation, and the 1990 step dummy. We also verify an inverse U-shaped relationship between international reserves and development level. *De facto* flexible exchange rate regime helps to explain reserve hoarding. But surprisingly, countries with flexible exchange rate regimes have higher ratios of reserves to GDP. This puzzling result is robust to alternative specifications.

The policy implications of our findings suggest a relationship between foreign reserves, financial flows globalization, and exchange rate regimes. Traditionally, it was thought that exchange rate flexibility was a substitute for reserve accumulation. However, we have seen that when countries are in an intermediate stage of development and face increasing exposure to capital flows, reserve accumulation can be considered an important tool to obtain a successful integration. In this scenario, flexibility could be viewed as a complement of reserves. This attitude is highly rational at a domestic level if we consider that we live in a world where there is no clear lender of last resort to deal with frequent financial and currency crises.

Appendix: Construction of the Variables and Data Sources

The dependent variable is the log ratio of international reserves to GDP. For international reserves the series Total Reserves “1..SZF” from the *International Financial Statistics* (IFS) was used. This includes the sum of the holdings of foreign currency, SDRs, gold, and the position at the IMF. To express it in dollars, the SDR conversion rate at the end of each period was employed (series “111..SA.ZF”). Gross domestic product is from the *World Development Indicators* (WDI) (specifically the “NY.GDP.MKTP.CD” series). What follows is a detailed explanation of the reserve demand determinants.

- GDP PPP per capita: it refers to the GDP PPP per capita at constant prices for 2000 series (“NY.GDP.PCAP.PP.KD” series of the WDI).

- Trade openness: this ratio was constructed using the total reserves series previously described and the sum of imports and exports of goods and services in current dollars ("NE.IMP.GNFS.CD" and "NE.EXP.GNFS.CD" lines in the WDI).
- Financial openness: this variable is the ratio of capital inflows and GDP "NY.GDP.MKTP.CD" series from WDI. Capital inflows series is constructed by adding the absolute value for the three items reported by the IFS: foreign direct investment (".78BEDZF" line) portfolio investment (".78BGDZF" line), and other investments (".78BIDZF" line). Only observations with data for the three items were considered.
- Volatility of exports: the above-mentioned export series was used in the construction of this variable. The volatility of year t was constructed taking the coefficient of variation for the exports of the years: t , $t-1$, $t-2$, $t-3$, $t-4$ via a rolling-window calculation.
- Volatility of capital inflows: capital inflow volatility (as in the financial openness variable) was calculated in the same way as its equivalent for exports.
- Opportunity cost: it is the annual yield on 10-year US Treasury bonds from the IFS ("111.61...ZF" series).
- Regional imitation: this variable calculates the ratio between the number of countries in a region that increased their total reserves during the previous year and the total number of countries making up the corresponding geographical region.
- Exchange rate regime dummies: the exchange rate regime for each country was classified each year following the methodology of Coudert and Dubert (2005). The regressions include the binary variables corresponding to the flexible and inconclusive categories, the omitted case being the exchange rate pegs. The considered regions are: East Asia and Pacific, South Asia, Europe and Central Asia, Latin America and Caribbean, North America, Middle East and North Africa, and Sub-Saharan Africa.
- Shift dummy variable: the 1990 shift binary variable takes the value 1 from 1990 to the end of the sample.

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Notes

1. Calculations are based on *International Financial Statistics* (IMF). Reserves in constant dollars were obtained by deflating the export price index of the United States.
2. An endogenous variable in the sense that $E(x_{it}, \varepsilon_{j,s}) = 0$ for all i, t, j, s .
3. The error term satisfies the following conditions: (i) $E(\varepsilon_{it}, \varepsilon_{j,s}) = 0, i \neq j$ or $t \neq s$, which states that error terms are mutually uncorrelated; and (ii) $E(\eta_i, \varepsilon_{j,s}) = 0$ for all i, j, s , i.e. errors are uncorrelated with the unobserved heterogeneity.
4. The differences GMM estimator of Arellano and Bond (1991) exploits linear moment conditions of the form $E(y_{i,t-s}\Delta\varepsilon_{it}) = 0$ for $t = 3, \dots, T$ and $2 \leq s \leq t-1$, and also allows incorporation of weakly endogenous covariates which are instrumented by the lagged levels dated at $t-1$. This estimator is consistent for $N \rightarrow \infty$ and fixed T .
5. Blundell et al. (2000) demonstrate that this condition is satisfied under mean stationarity of the y_{it} process. The result can be extended to the endogeneity case of the x_{it} process.
6. Traditional literature emphasized the relevance of the propensity to import as a measure of the costs of an external adjustment. As a consequence, a negative relationship was expected between reserves and propensity to import. However, in empirical research a positive association was systematically found which was attributed to the fact that the latter is a proxy of trade openness. In line with this interpretation, we chose the standard measure of trade openness.
7. This point has been modeled by Aizenman and Lee (2006) employing a game-theory setting. They explain the hazard of competitive accumulation emphasizing the existence of negative externalities comparable to those of the competitive devaluation.
8. We use the two-step variant of the GMM model incorporating the finite sample variance correction of Windmeijer (2005). The two-step GMM estimator is efficient but the estimated asymptotic standard errors could be severely downward biased in small samples. For this reason, common practice was to sacrifice some precision (using the one-step GMM estimator) in order to avoid large inference mistakes. Windmeijer (2005) demonstrates that the presence of estimated parameters on the weight matrix accounts for much of the divergence between a finite sample and the usual asymptotic variance of the two-step GMM estimator. This divergence could be estimated resulting in a finite-sample-corrected variance that approximates fairly well the sample variance and thus is conducive to a more accurate inference.
9. In GMM models the number of instruments grows exponentially when T increases. This implies that in a typical macro panel data (larger T and smaller N than in a micro panel) it is common for the second step variance-covariance matrix to become singular if instruments are not restricted. This usually weakens the Hansen test to the point of generating unlikely good p -values of one. In addition, a large instrument collection could overfit endogenous variables. As a practical rule of thumb it is recommended that the number of instruments do not exceed the number of cross-section units (Roodman, 2006). Using the collapse option in the `xtabond2` routine of STATA (Roodman, 2006) allows us to not surpass this limit.
10. For instance, Kenen and Yudin (1965), Heller (1966), Clark (1970), or Frenkel and Jovanovic (1981).
11. This reason was previously used in Soto et al. (2004) to explain the counterintuitive negative effect of the opportunity cost (interest rate differential) over international reserve holdings.
12. We are grateful to Thomas Wu and an anonymous referee to suggest this line of exposition for this result.
13. The exception are the dummy variables. In this exercise both the dummy variable of the flexible exchange rate regimes and the dummy for the 1990–2003 period were set equal to one.